

Amr Ashraf Amin Moussa

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Profile

Data Scientist and AI Agent Developer with hands-on experience building end-to-end machine learning pipelines, LLM-powered applications, and Retrieval-Augmented Generation (RAG) agents using Python, SQL, LangChain, and TensorFlow. Skilled in data cleaning, exploratory data analysis (EDA), feature engineering, and model deployment, with practical MLOps experience (MLflow, DVC, Docker). Proven ability to design autonomous AI agents and conversational systems backed by vector databases and LLM APIs (Groq, HuggingFace). Highly motivated to contribute to data-driven and AI agent engineering teams.

Education

Helwan University Cairo, Egypt
Bachelor of Science (B.Sc.) – Computer Science & Statistics | CGPA: 3.769 2023 – 2027

Professional Experience

Orange Digital-Hub Dokki, Egypt
AI Agent Trainee 07/2026 – 08/2026

- Completed an intensive training program on AI agents and the LLM ecosystem, covering LLM fundamentals and prompt engineering techniques.
- Built chains and tools using LangChain, and implemented memory and context handling for conversational agents.
- Integrated external knowledge sources into agent workflows and designed multi-agent collaboration systems.
- Added automation and API integrations to extend agent capabilities, and built a web interface to interact with the agents.
- Performed evaluation, debugging, and optimization of AI agents, culminating in a final presentation and deployment of the project.

Digital Egypt Pioneers Initiative (DEPI) Cairo, Egypt
Data Scientist Intern 06/2025 – 12/2025

- Worked on real-world data science projects using Python and SQL as part of the DEPI internship program.
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) to extract actionable insights from raw datasets.
- Created data visualizations to communicate findings clearly and support data-driven decision-making.
- Applied machine learning techniques for prediction, forecasting, and analysis tasks.
- Collaborated with team members in a data-driven, project-based environment.

Projects

AI Medical Assistant – RAG-Based Conversational Agent | *Python, LangChain, FAISS, HuggingFace, Groq API, Jinja2*

- Built an autonomous AI agent for medical Q&A using a Retrieval-Augmented Generation (RAG) pipeline with LangChain, a FAISS vector store, and HuggingFace embeddings.
- Migrated the LLM backend to the Groq API for faster inference and refactored the prototype into a modular, production-ready Streamlit application with a text-to-pandas analytics layer.
- Delivered production-readiness improvements including Docker and docker-compose deployment, FAISS index persistence, and multi-session chat history management.

Store Sales – Time Series Forecasting | *Python, Pandas, Matplotlib, Seaborn, DVC, MLflow* 07/2025 – 11/2025

- Developed an end-to-end solution for forecasting grocery sales for Corporacin Favorita (Kaggle competition).
- Built a robust preprocessing pipeline handling holidays, oil prices, and transaction data.
- Implemented a multi-model forecasting system using Prophet, ARIMA, and LightGBM with automated model selection.
- Designed interactive dashboards with Streamlit for real-time sales prediction and scenario analysis.
- Applied MLOps practices including data and model versioning with DVC and Git, experiment tracking, and model registry with MLflow.
- Implemented model monitoring with drift detection, performance metrics (MAE/RMSE), and automated alerts for retraining.

Cairo Real Estate Pricing | *Python, Streamlit, Scikit-Learn, Gradient Boosting*

- Developed a Streamlit application to predict real estate prices in Cairo based on numeric and categorical property features.
- Performed EDA and feature engineering on Cairo housing data; applied gradient boosting for accurate price estimation.

Certificates

Machine Learning with Python – IBM, Coursera

- Completed IBM's professional certificate covering supervised and unsupervised learning, regression, classification, and clustering.
- Applied machine learning algorithms using Python libraries including Scikit-Learn, SciPy, and NumPy on real-world datasets.

Certified Data Scientist – DEPI

- Completed an intensive training program covering data analysis, Python, SQL, and machine learning fundamentals through real-world projects.

Python Programming Basics – Mahara Tech

- Covered Python fundamentals including data types, control flow, functions, and basic problem-solving.

SQL Certificate – HackerRank

- Gained practical experience in writing SQL queries, data retrieval, filtering, aggregation, and joins.

Python (Basic) – HackerRank

- Validated core Python proficiency through timed algorithmic challenges, demonstrating logic and syntax mastery.

Skills

Programming: Python, SQL, Shell Scripting

AI Agents & LLMs: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM APIs (Groq, HuggingFace), Vector Databases (FAISS), Conversational AI Agents

Data Science & ML: Data Analysis, Machine Learning, Deep Learning, Time Series Analysis, Feature Engineering, EDA, Statistical Analysis

Frameworks & Tools: TensorFlow, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Prophet, LightGBM

MLOps & Deployment: MLflow, DVC, Docker, Git & GitHub, Model Monitoring, Experiment Tracking, REST APIs

Other: Data Visualization & Dashboards, Linux, Big Data (Theoretical)

Soft Skills: Problem Solving, Communication, Collaboration & Teamwork, Time Management, Adaptability & Learning

Languages

Arabic: Native

English: Proficient